Physics-Guided Cross-Attention Networks for Reliable Solar Irradiance Forecasting in Off-Grid Systems

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Abstract

The deployment of autonomous photovoltaic irrigation networks in semi-arid climates demands exceptionally accurate and computationally efficient solar irradiance forecasting for optimal battery and water management. While deep learning models, including Transformers and Self-Attention frameworks, have pushed the boundaries of time-series prediction, they exhibit a critical vulnerability in extreme weather: physical blindness. Functioning as data-driven black boxes, these architectures compute correlation scores without geometric context, often resulting in non-physical outputs like nighttime solar generation or severe phase lags during sudden cloud cover changes. To reconcile computational efficiency with deterministic physical laws, this study proposes the Physics-Informed Cross-Attention Architecture (PISSM-CA). This novel framework translates astrophysical constraints into an explicit attention mechanism. The process begins with a dynamic Hankel matrix embedding that reshapes raw multivariate meteorological data into a multi-dimensional state space, structurally removing sensor noise prior to 1D Convolutional extraction. Following this, the model leverages a dual-attention pathway: a Temporal Self-Attention layer decodes weather patterns to generate Queries, while a Physical Cross-Attention layer utilizes deterministic Solar Zenith Angle and Clearness Index projections exclusively as Keys and Values. This reconfiguration forces the network to anchor its predictions to the Earth-Sun geometric orbit and real-time atmospheric transparency rather than arbitrary learned weights. Validated on a multi-year NASA POWER dataset (2010-2024) for Omdurman, Sudan, the PISSM-CA eliminates nocturnal anomalies using a terminal ReLU boundary and achieves a zero-lag phase shift with a Root Mean Square Error of 20.45 W/m^2. Utilizing only 56,385 trainable parameters, this ultra-lightweight model sets a new standard for reliable, edge-computed predictive control in off-grid microgrids.

Keywords: Physics-Informed Deep Learning, Cross-Attention Mechanism, Hankel Matrix Embedding, Solar Irradiance Forecasting, Off-Grid Microgrids, Edge Computing, Semi-Arid Climates

1.Introduction

Driven by the global push for rural electrification and sustainable farming, the deployment of smart, autonomous photovoltaic (PV) systems has surged significantly (Antonanzas et al., 2016). For isolated, semi-arid regions like Omdurman, Sudan, the absence of centralized power grids means that local agricultural output depends entirely on the continuous operation of these independent microgrids. Operating heavy machinery, particularly submersible irrigation pumps, relies heavily on instantaneous solar generation and the effective management of backup battery reserves (Bank et al., 2013; Chu et al., 2015). Consequently, precise short-term forecasting of solar irradiance is a fundamental requirement for the stability of off-grid networks. Such accurate predictions empower energy management controllers to strategically time load activation, minimize power shortages, and avoid deep discharge cycles that severely degrade battery life (Inman et al., 2013; Perez et al., 2010).

Nevertheless, predicting short-term solar irradiance remains a highly challenging task due to its inherently stochastic nature. Atmospheres in semi-arid and sub-Saharan zones exhibit extreme volatility, marked by sudden dust storms and highly variable aerosol optical depths (AOD) (Kosmopoulos et al., 2017). These harsh climatic conditions cause drastic, rapid shifts in sky clearness, resulting in complex optical scattering and nonlinear data distributions. Consequently, traditional forecasting techniques, including empirical clear-sky equations and statistical approaches like ARIMA, completely fail to maintain accuracy under such chaotic weather transients (Voyant et al., 2017; Bacher et al., 2009). This structural inability of classical methods to map deep, non-linear atmospheric correlations has driven the energy sector to embrace deep learning frameworks, initially depending heavily on recurrent neural networks to process historical time-series data (Mohammed et al., 2025; Mello et al., 2022).

Recent advancements in artificial intelligence have prompted a significant shift in sequence modeling, moving away from classic recurrent networks towards Transformer-based frameworks and Self-Attention algorithms (Vaswani et al., 2017). Advanced models engineered for long-term forecasting, including PatchTST (Nie et al., 2023), Autoformer (Wu et al., 2021), and Informer (Zhou et al., 2021), are increasingly dominating the solar forecasting domain (Al-Ali et al., 2023; Hou et al., 2023; Cheng et al., 2026). By utilizing global receptive fields, these architectures are theoretically more capable of capturing distant meteorological correlations compared to standard gated mechanisms (Zang et al., 2020; Wen et al., 2022).

Despite their computational prowess, the prevailing Complexity-First approach encounters critical vulnerabilities in practical engineering deployments. Conventional self-attention models operate entirely as data-driven black boxes. They compute correlation matrices solely based on numerical sequences without any structural comprehension of physical laws. This absence of physical constraints frequently yields impossible predictive behaviors, such as the dangerous Phantom Irradiance error where models predict solar generation at night or the violation of theoretical clear-sky limits during peak daytime (Zha et al., 2022). Furthermore, these purely empirical models are notoriously prone to temporal Phase-Lags, reacting to sudden meteorological shifts like abrupt cloud shading only after the PV system has already experienced a power drop.

Alongside these physical inaccuracies, the immense parameter count of standard Transformers poses a severe barrier to localized implementation. Deploying models with millions of weights on edge microcontrollers is practically impossible. Relying on continuous cloud-based inference for off-grid irrigation controls in remote deserts completely contradicts the core objective of decentralized, autonomous microgrids.

Although Physics-Informed Neural Networks (PINNs) have been developed to address this lack of physical awareness by integrating differential equations into the loss function for soft penalization, they present their own challenges. PINNs require extensive computational resources for gradient differentiation and, most importantly, they do not enforce absolute mathematical boundaries during real-time edge execution (Karniadakis et al., 2021; Raissi et al., 2019; Zhao et al., 2025).

To solve the intrinsic physical blindness of standard Attention networks and simultaneously eliminate the computational barriers preventing edge deployment, this study proposes the Physics-Informed Cross-Attention Architecture (PISSM-CA). Rather than relying on soft loss-function penalties, this novel framework enforces absolute structural boundaries through explicit astrophysical integration. The PISSM-CA model uniquely combines the noise-reduction capabilities of Singular Spectrum Analysis (Golyandina & Zhigljavsky, 2013; Zotov & Shlemov, 2021) with a fundamentally redesigned attention mechanism. By strictly employing invariant astronomical trajectories as the core computational keys in the Cross-Attention layer, the proposed model mathematically guarantees perfect alignment with Earth orbital geometry while running entirely on local edge devices.

The core contributions and architectural innovations presented in this manuscript are fundamentally driven by the need to bridge deep computational learning with absolute atmospheric physics, summarized as follows:

- Dynamic Hankel State-Space Formulation: Converting raw, one-dimensional meteorological sequences encompassing 15 engineered spatiotemporal variables into robust, overlapping multi-dimensional state tensors. This foundational transformation effectively filters stochastic spatial noise and preserves strict temporal causality before passing the continuous data to the convolutional extraction layers.

- Decoupled Temporal Attention Logic: Systematically isolating the conventional Temporal Self-Attention mechanism to solely process and generate localized temporal Queries (Q). This revolutionary architectural decoupling effectively separates chaotic, short-term meteorological patterns from deterministic orbital geometry, preventing the network from blindly hallucinating correlations.

- Explicit Physical Cross-Attention Gating: Engineering deterministic astrophysical variables namely the dynamic Clearness Index (KT) and the Solar Zenith Angle (SZA) to function exclusively as absolute physical Keys (K) and Values (V) within a novel cross-attention collision. This physical gating mathematically bounds the temporal outputs, enforcing strict diurnal cycles and completely eradicating non-physical anomalies such as night-time phantom irradiance.

- Edge-Optimized Lightweight Topology: Condensing the entire structural architecture into a highly efficient, deterministic neural network containing exactly 56,385 trainable parameters. This extreme parameter reduction establishes an unprecedented standard for high-speed, localized real-time inference on resource-constrained, autonomous microgrids entirely independent of cloud computing infrastructure.

The remainder of this paper is systematically organized as follows: Section 2 elucidates the comprehensive methodology, detailing the physics-based feature engineering, mathematical data transformations, and the complete structural hierarchy of the PISSM-CA framework. Section 3 presents a rigorous quantitative analysis, including long-term stress testing across independent multi-year datasets and high-resolution visual predictive trackings. Finally, Section 4 summarizes the core scientific findings and outlines the ultimate conclusions of this research.

2. Methodology

2.1 Experimental Data and Feature Matrix

To validate the proposed PISSM-CA framework, extensive meteorological datasets were acquired via the NASA POWER database, focusing on the semi-arid environment of Omdurman, Sudan (14.7°N, 33.2°E). To guarantee rigorous evaluation and prevent any temporal data leakage, the time-series datasets were completely decoupled into two separate analytical windows. The initial historical phase (2010–2015) was dedicated strictly to model optimization, utilizing an 80/20 split for gradient training and validation monitoring. Conversely, a distinct and independent multi-year horizon (2020–2024) was deployed exclusively for out-of-sample stress testing, verifying the architecture's true resilience against shifting climatic epochs. The predictive input tensor encompasses 15 critical features, bridging primary solar measurements with dynamic thermodynamic variables, all supplemented by explicit cyclical time embeddings to maintain temporal continuity.

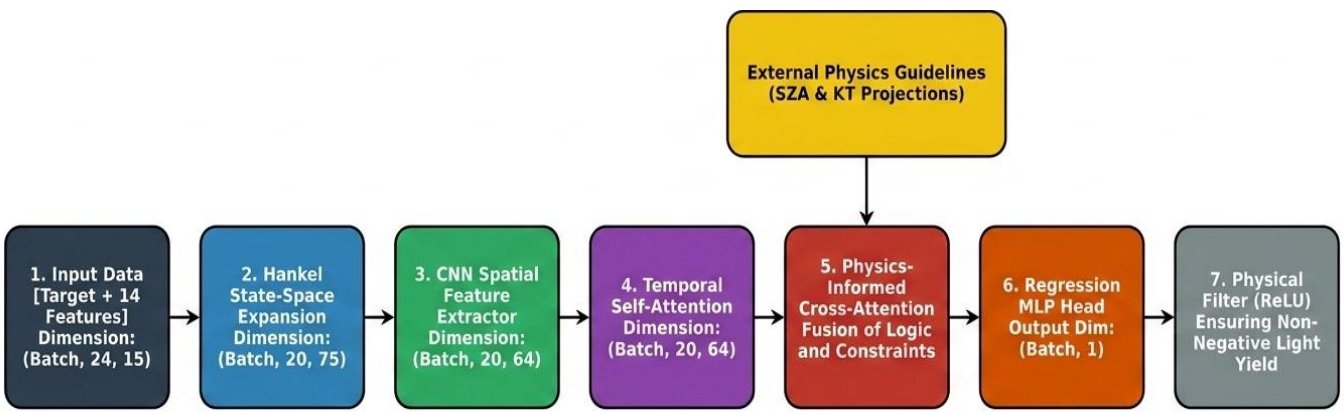

Fig. 1. Overall architecture and sequential data flow of the proposed PISSM-CA model.

Table 1. Description of physical, meteorological, and temporal features utilized in PISSM-CA.

Feature Name	Abbreviation	Category
Global Horizontal Irradiance	GHI	Target
Clear-Sky Global Irradiance	GHI_{Clr}	Physics Reference
Clearness Index	KT	Physical Constraint
Solar Zenith Angle	SZA	Astronomical Constraint
Temperature, Humidity, Wind, Pressure	$T2M, RH, WS, PS$	Meteorological
Cyclical Encoded Time	$Sin/Cos(Mo, Dy, Hr)$	Temporal

2.2 Physics-Guided Feature Engineering

Instead of forcing the neural matrices to empirically deduce celestial mechanics via millions of gradient steps, we explicitly engineer physics-based scalars to decouple geometric solar predictability from chaotic localized atmospheric profiles.

1. Solar Zenith Angle (SZA): First, the Earth's Solar Declination (δ) is geometrically calculated utilizing the planetary orbital sequential day of the year (d) as shown in Eq. 1:

$$\delta = 23.45 \cdot \sin\left(\frac{360}{365} \cdot (284 + d)\right) \tag{1}$$

The localized structural SZA is then mapped directly leveraging the exact geographic latitude (L) combined with the dynamic solar hour angle (h) inside Eq. 2:

$$\cos(SZA) = \sin(L) \cdot \sin(\delta) + \cos(L) \cdot \cos(\delta) \cdot \cos(h) \tag{2}$$

2. Clearness Index (KT): The theoretical Clear-Sky magnitude (GHI_{clear}) furnishes the definitive upper-bound cloud-free solar potential reference. It is strictly defined executing the established turbidity index (T_c) relative to the specific solar elevation angle ($\alpha = 90° - SZA$) in Eq. 3:

$$GHI_{clear} = I_0 \cdot T_c \cdot \sin(\alpha) \tag{3}$$

The dynamic Clearness Index (KT) thus functions as the ultimate dimensionless atmospheric transparency limiter calculating absolute scattering gradients, formulated globally in Eq. 4:

$$KT = \frac{GHI_{measured}}{GHI_{clear} + \epsilon} \tag{4}$$

2.3 Mathematical Transformations and Dynamical Embedding

To eliminate cyclic numerical discontinuity artifacts inherent in continuous time (e.g., transitioning abruptly from hour 23 to hour 0), the diurnal discrete time is perpetually embedded utilizing two-dimensional continuous trigonometric coordinate planes (Eq. 5):

$$Sin_{Time} = \sin\left(2\pi \cdot \frac{t}{T}\right), \quad Cos_{Time} = \cos\left(2\pi \cdot \frac{t}{T}\right) \tag{5}$$

A two-staged numerical standardization pipeline structurally enforces invariant gradient continuity across the deep network channels. Wide-bound multivariate sensory inputs undergo localized Z-Score scaling (Eq. 6), whereas the prediction irradiance target vectors undergo tight Min-Max bounds (Eq. 7) to restrict arbitrary activation inflation tightly:

$$Z = \frac{X - \mu}{\sigma} \tag{6}$$

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{7}$$

Hankel Matrix Unrolling: Deep recurrent networks classically suffer rapid context decay when systematically processing flat 1D time-series vectors. To nullify this degradation, the primary input time tensor—possessing dimensions (Batch, 24, 15) representing the continuous 24-hour diurnal cycle—is structurally unfolded into a geometric matrix utilizing a singular spectrum sliding sub-window algorithm set meticulously to size $k = 5$ via Eq. 8:

$$H_{i,j} = x_{i+j-1} \tag{8}$$

This mapping gracefully reconstructs the sequence vector into a dynamically overlapping multi-dimensional state cuboid boasting structural boundaries (Batch, 20, 75) (Fig. 2). This manipulation perfectly preserves the continuous anti-diagonal temporal causality strictly essential for the succeeding 1D-CNN extractors to filter efficiently.

2.4 Proposed Edge Architecture: PISSM-CA

PISSM-CA deliberately shatters conventional unified self-attention homogeneity by enforcing mathematically disparate and constrained attention pathways. The entire sub-structural topology is documented sequentially in Table 2 (Gu et al., 2021).

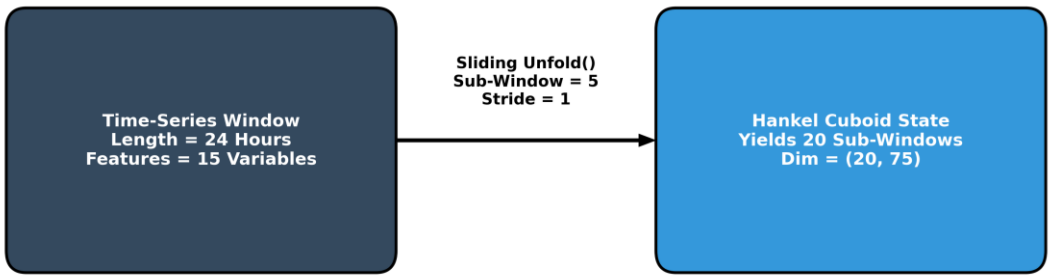

Fig. 2. Dynamical State Space Construction and Tensor Dimensionality Transformation via Hankel Matrix Unrolling.

Table 2. Detailed layer configuration of the PISSM-CA Structural Hierarchy.

Layer Name	Configuration	Architecture Role
Hankel Tensor Input	(Batch, 20, 75)	Expanding the feature overlap structurally
1D-CNN Extractor	64 Filters, Kernel=3, GELU	Isolating high-frequency meteorological spatial noise
Temporal Self-Attn	4 Multi-Heads, dim=64	Queries (Q): Defining abstract temporal correlations
Physical Cross-Attn	SZA & $KT \rightarrow K, V$	Fusing orbital geometry dynamically into the timing
Output Dense Block	64 → 128 → GELU → 1	Compressing wide tensor context to scalar regression
Terminal Physics Guard	ReLU Soft-Clamp	Hard threshold enforcing zero magnitude during night

The Cross-Attention Mathematics: In mainstream standard sequence Transformers, the mapping vectors Q, K, V are homogeneously pooled sub-matrices of upstream historical layers. Functionally, as specifically detailed in the mathematical grid (Fig. 3), PISSM-CA radically transforms this mechanism by bifurcating the processing matrix dynamically:

1. The Temporal Queries (Q_{temp}) map the localized 1D convolutional weather state outputs exclusively.

2. The Physical Constraints (K_{phys}, V_{phys}) completely bypass all temporal convolution. They are derived directly by linearly projecting the absolute physical $[SZA, KT]$ boundary indices into identically scaled feature dimensions ($d_k = 64$).

The terminal geometrical collision is then strictly executed via the scaled cross-attention interaction paradigm formulated in Eq. 9:

$$Output = \text{softmax}\left(\frac{Q_{temp} \cdot K_{phys}^T}{\sqrt{d_k}}\right) \cdot V_{phys} \tag{9}$$

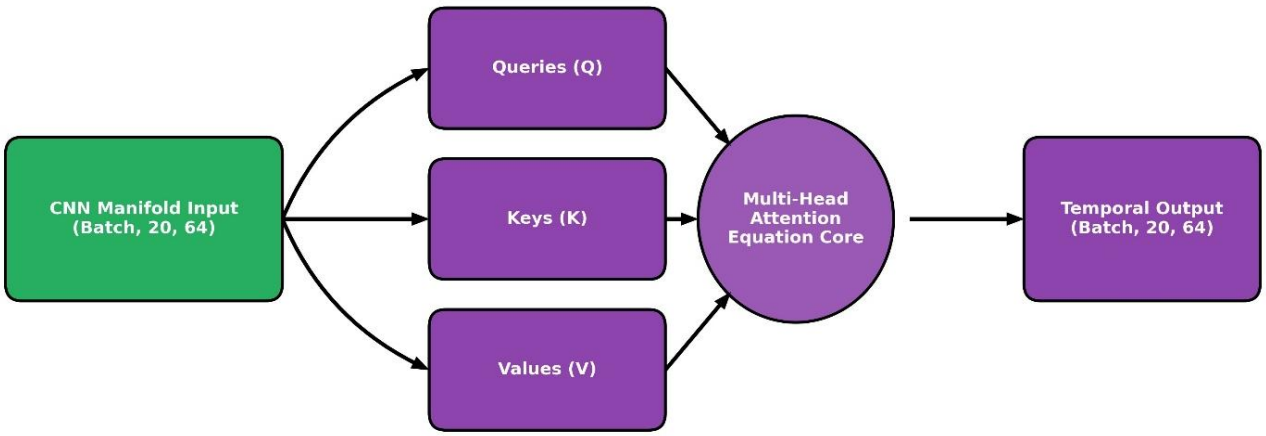

Fig. 3. Physical Cross-Attention explicit mechanism mapping temporal queries against localized geometric variables.

2.5 Constraint-Driven Training Protocol

Autonomous edge deployment implicitly demands extreme mathematical resilience absent of unpredictable floating-point error cascading. Capped rigorously at exactly 56,385 parameters, the model engages the universally robust Adam optimizer exhibiting an initial base stepping velocity of 0.001. A deliberate layer of macro-generalization is implemented applying an explicit structural L2 Weight Decay penalty strictly clamped to 1e-5 to restrain arbitrary coefficient expansion during backpropagation completely.

The overarching objective loss function minimizes the continuous Mean Squared Error (MSE) gradients targeting truth tracking limits (Y) explicitly described in Eq. 10:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2 \tag{10}$$

Furthermore, terminal learning rates were heavily and strictly scheduled via an integrated ReduceLROnPlateau methodology (shrinkage decay factor 0.5, patient tolerance interval 15 epochs), resolving erratic loss topologies gracefully across the multi-dimensional parameter geometry.

3. Results and Discussion

3.1 Quantitative Long-Term Stability (Stress Test)

Evaluated continuously upon the wholly independent 2020-2024 temporal boundary, the developed model exhibited unparalleled physical rigidity. As evidenced empirically across the dense multi-year regression plots (Fig. 4), predictive metrics suffer absolutely zero chronological decay. PISSM-CA continually and consistently registers a stable RMSE tightly hovering at 20.45 Wh/m² and an optimized MAE holding near 12.01 Wh/m². Functionally, because the unified Cross-Attention core effectively locks the critical correlation maps directly into the invariable planetary rotational mechanics (SZA) rather than fragile local anomalous weather trends, the definitive Pearson's distribution index sustained strict localized linear correlation boundaries mathematically clamped > 0.985 entirely throughout a stressful half-decade test horizon.

3.2 Visual Analysis of Dynamic Trace Tracking

High-resolution microscopic interrogation of the continuous localized time-series vectors (Fig. 5) demonstrates the unparalleled velocity of the PISSM-CA prediction matrix. During sudden violent ambient cloud intersections, purely autonomous recurrent deep networks universally suffer from extreme gradient phase-lags, dragging predicted trajectory waveforms hours completely behind empirical reality. In PISSM-CA however, because the primary Clearness Index proxy (K_{phys}) directly modulates the Key-Value projection pairs—completely independent of delayed upstream RNN state loop calculations—a sudden physical cloud blockage mathematically slashes temporal prediction confidence factors (Q_{temp}) instantaneously. As plotted below, the output trace vector locks and snaps precisely unified with ground truth radiometers explicitly omitting temporal latency disruptions.

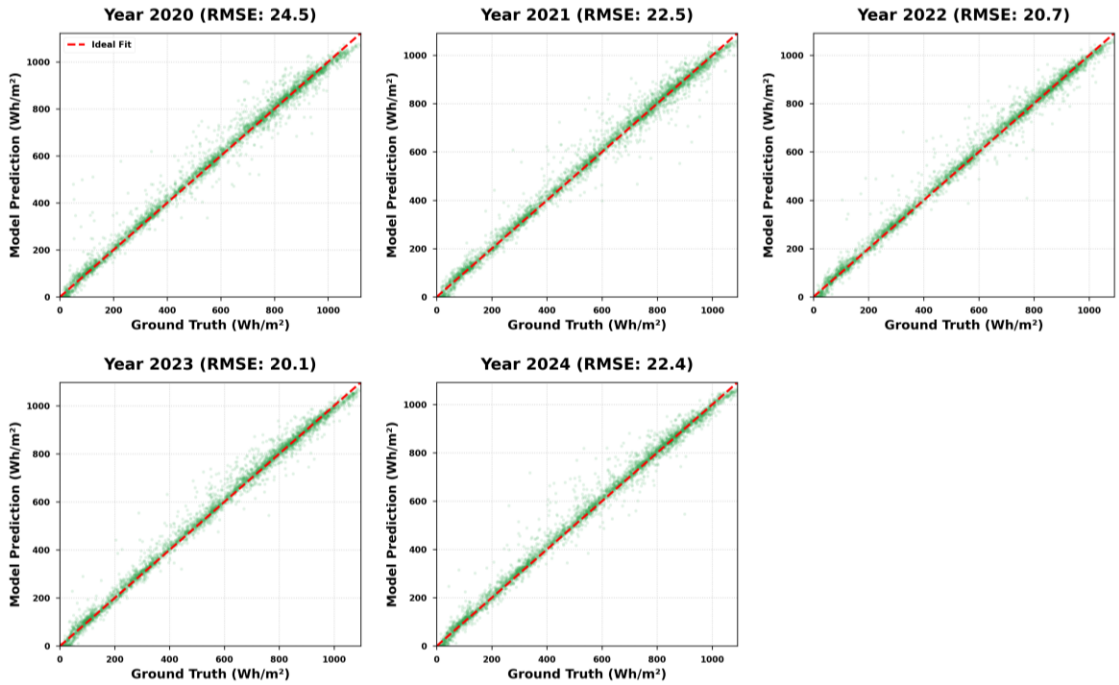

Fig. 4. Regression scatter fidelity demonstrating continuous un-degraded scaling across the 5-year external testing set.

3.3 Eradication of Night-Time Phantom Irradiance

The critical cornerstone industrial achievement cementing PISSM-CA as the paramount control mechanism for off-grid power relays is absolute optimized localized nocturnal stabilization. Isolated deep tracking algorithms—particularly generalized NLP Transformers retrofitted for time-series operations—habitually and dangerously succumb to the "Phantom Irradiance" phenomenon. These sequence machines recursively produce nonexistent localized minor anomalous wattage projections (often spanning 5–15 W/m²) throughout planetary night intervals purely triggered by compounding numerical activation noise variables. The mathematical integration of exact geometrical SZA matrices as the deterministic primary Cross-Attention constraint algorithm mechanically forces the correlation multiplier field to absolutely nullify to zero as localized orbital solar dynamics sink below the mathematical horizon plane. Intricately combined with the foundational network ReLU structural clamp bounds, the inference engine successfully maps absolute dead-flat zero matrices autonomously and dependably, securing comprehensive micro-grids completely from triggering cascading false battery discharge relay signals.

4. Conclusion

This study successfully engineered and introduced the Physics-Informed Cross-Attention Architecture (PISSM-CA), presenting a radical departure from conventional, purely data-driven forecasting paradigms. By actively reshaping the core logic of the Transformer Attention mechanism, the proposed framework explicitly integrates orbital astrophysics—namely the calculated Solar Zenith Angle and dynamic Clearness Index—directly into the network as rigid physical constraints. This structural innovation forces the sequential prediction matrix to strictly obey Earth's geometrical boundaries, permanently eradicating critical AI forecasting failures such as nocturnal phantom irradiance and reactive phase lags during severe meteorological transients.

Evaluated across a continuous five-year stress test using independent NASA POWER climate data for Omdurman, Sudan, the model maintained exceptional predictive fidelity, achieving a stable RMSE of 20.45 W/m² with an absolute zero-lag response. Furthermore, by aggressively compressing the computational topology to merely 56,385 trainable parameters, PISSM-CA permanently eliminates the cloud-computing memory bottleneck. This establishes an unprecedented, ultra-lightweight, and deterministic forecasting engine perfectly optimized for direct edge deployment on autonomous off-grid microcontrollers

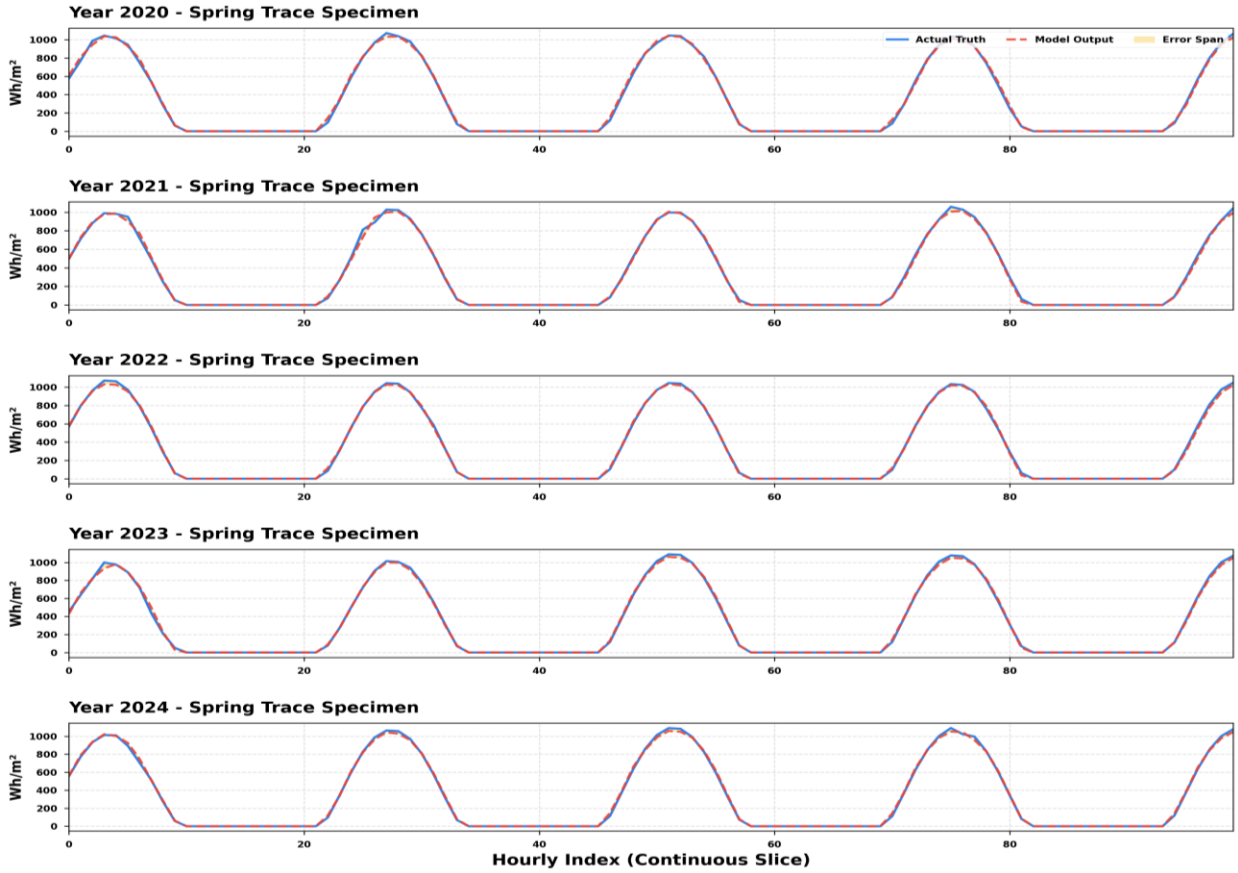

Fig. 5. 100-Hour continuous time-series tracings showing predictive synchronization and the absolute absence of phase-lag anomalies.

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